AECCS: Assessing the Effectiveness of Cookie Consent Systems

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CS475 Privacy-Enhancing Technologies · Final report template aligned with the current repository state

Abstract

AECCS is an implementation-focused measurement system for studying cookie consent systems, GDPR-oriented compliance signals, dark patterns, and privacy-enhancing technologies (PETs). The repository now contains a provenance-aware pipeline that separates synthetic development data from the final real-study dataset. The current implementation supports automated crawling, tracker classification, dark-pattern detection, compliance scoring, browser PET evaluation, CMP comparison, differential privacy reporting, and report generation. The remaining gap is empirical rather than architectural: the real crawl and real PET experiments must be executed under the `data/real/` tree before final claims are made.

> The metrics currently available in the repository are demonstration values generated from the synthetic dataset in `data/mock/`. They are useful for validating the pipeline and report structure, but they are not final study findings.

1. Problem Statement

Cookie consent systems are intended to provide meaningful user control over non-essential tracking. Prior work and regulatory guidance suggest that many banners either manipulate users toward acceptance or fail to technically enforce rejection. The core question of this project is whether consent systems protect privacy in practice and, when they do not, whether PETs reduce user exposure.

2. Related Work

AECCS is grounded in three main strands of prior work: studies of dark patterns in consent banners, studies of enforcement failures after rejection, and large-scale automated cookie-consent measurement. A companion literature review is included in `docs/literature_review.md`.

Key references

- Nouwens et al., *Dark Patterns after the GDPR*
- Matte et al., *Do Cookie Banners Respect My Choice?*
- Bollinger et al., *Automating Cookie Consent and GDPR Violation Detection*
- GDPR Regulation (EU) 2016/679
- CNIL cookie and tracker guidance

3. System Overview

The implemented pipeline follows the proposal closely. Each target site is visited in three states: no interaction, accept all, and reject all. Raw crawl outputs are then processed into tracker classifications, dark-pattern findings, compliance scores, aggregate statistics, PET comparisons, and report artifacts.

Module	Status
Crawler	Implemented
Tracker classification	Implemented
Dark-pattern detection	Implemented
Compliance scoring	Implemented
Aggregate metrics	Implemented
Browser PET evaluation	Implemented, awaiting real runs
CMP analysis	Implemented
Differential privacy reporting	Implemented
Visualization and HTML reporting	Implemented

4. Data Hygiene and Reproducibility

A major repository milestone in the current phase was separating historical mixed outputs from the active datasets. New runs now target `data/mock/` or `data/real/`, while older root-level artifacts have been quarantined under `data/legacy/` and `reporting/legacy/`.

Raw and processed artifacts now carry provenance fields including source mode, run ID, generation timestamp, browser information, proxy state, and study-list origin. The report generator rejects mixed processed inputs to avoid blending mock and real findings.

5. Current Demo Findings

The synthetic dataset currently bundled with the repository contains 20 mock sites, of which 19 are successful crawls and 1 is a deliberate failure case. These values should be read as pipeline-validation outputs, not final empirical conclusions.

Metric	Current demo value
Pre-consent tracker violations	84.2%
Sites with any dark pattern	75.0%
Average compliance score	50.8 / 100
Best browser PET	Brave Shields (98.5% average reduction)
Best CMP	Didomi (PET score 75.1)
Recommended DP epsilon	2.0

6. Proposal Coverage

The original proposal committed to a real-site automated measurement study over 100 to 200 websites. The repository now includes a 100-domain seed list in `data/websites.csv`, which brings the project back in line with the proposal at the planning level. What remains is execution of the final study under EU-like browsing conditions and regeneration of all figures and report outputs from real data only.

7. Remaining Work

- Execute the final real crawl into `data/real/raw/`.
- Run the real PET subset or full PET evaluation into `data/real/processed/`.
- Regenerate aggregate metrics, DP outputs, CMP comparison, figures, and HTML report from real data only.
- Replace the demo findings in this document with real-study findings and citations.

8. Ethics and Limitations

The project uses automated browsing and therefore must remain rate-limited and transparent about its collection scope. The current repository includes synthetic data for development and validation, but no final claims should be made until the real-site runs are completed. The current draft also does not yet contain a final reference section formatted for an ACM submission.

9. Conclusion

AECCS is now structurally ready for an honest final course submission. The core software system is built, the data model is provenance-aware, and the mock dataset validates the end-to-end pipeline. The final step is operational: execute the real measurement study and replace the demonstration findings with real evidence.

References

Final ACM-format citations should be inserted here after the real-result draft is prepared. The literature review file already identifies the core sources to cite.